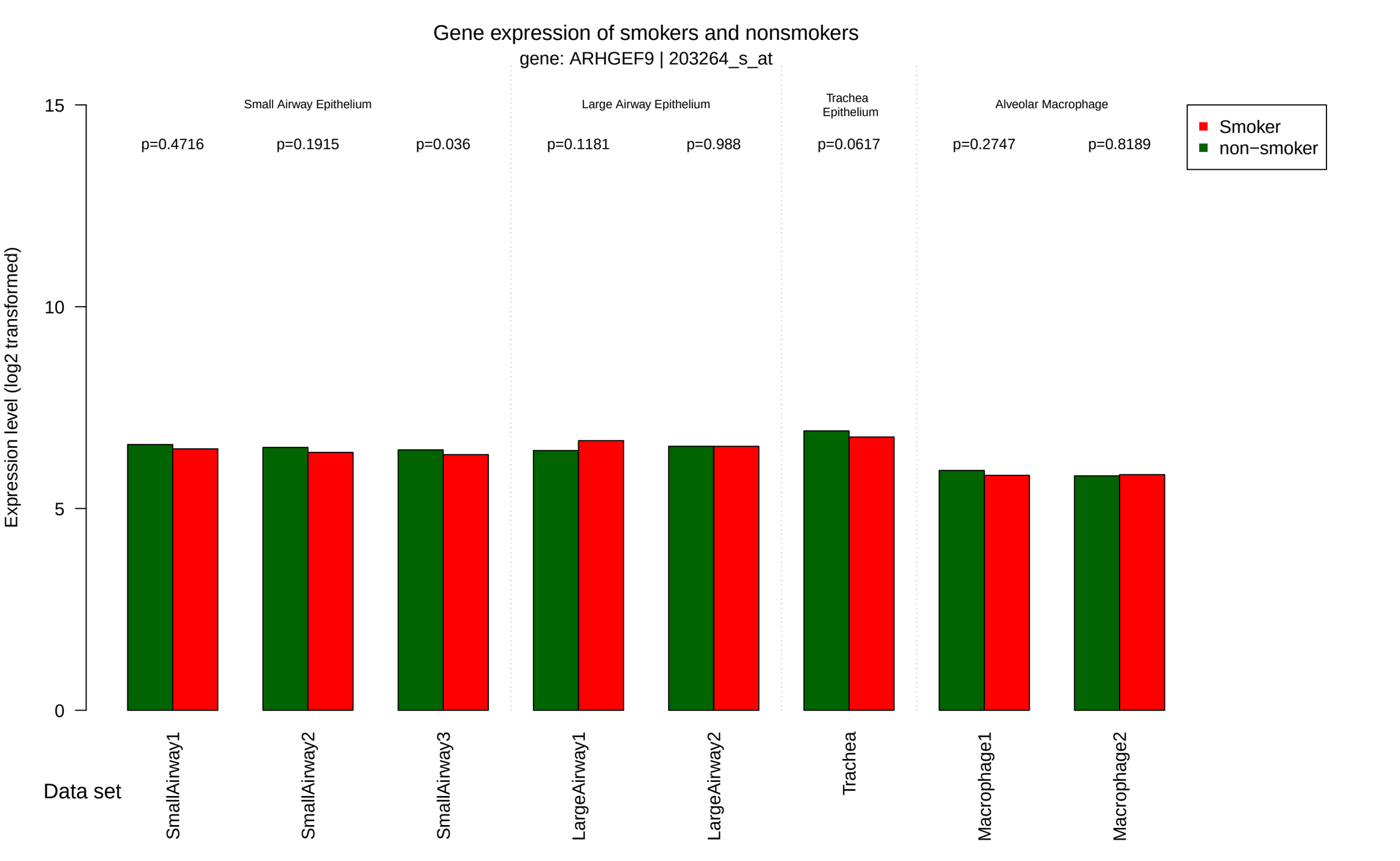
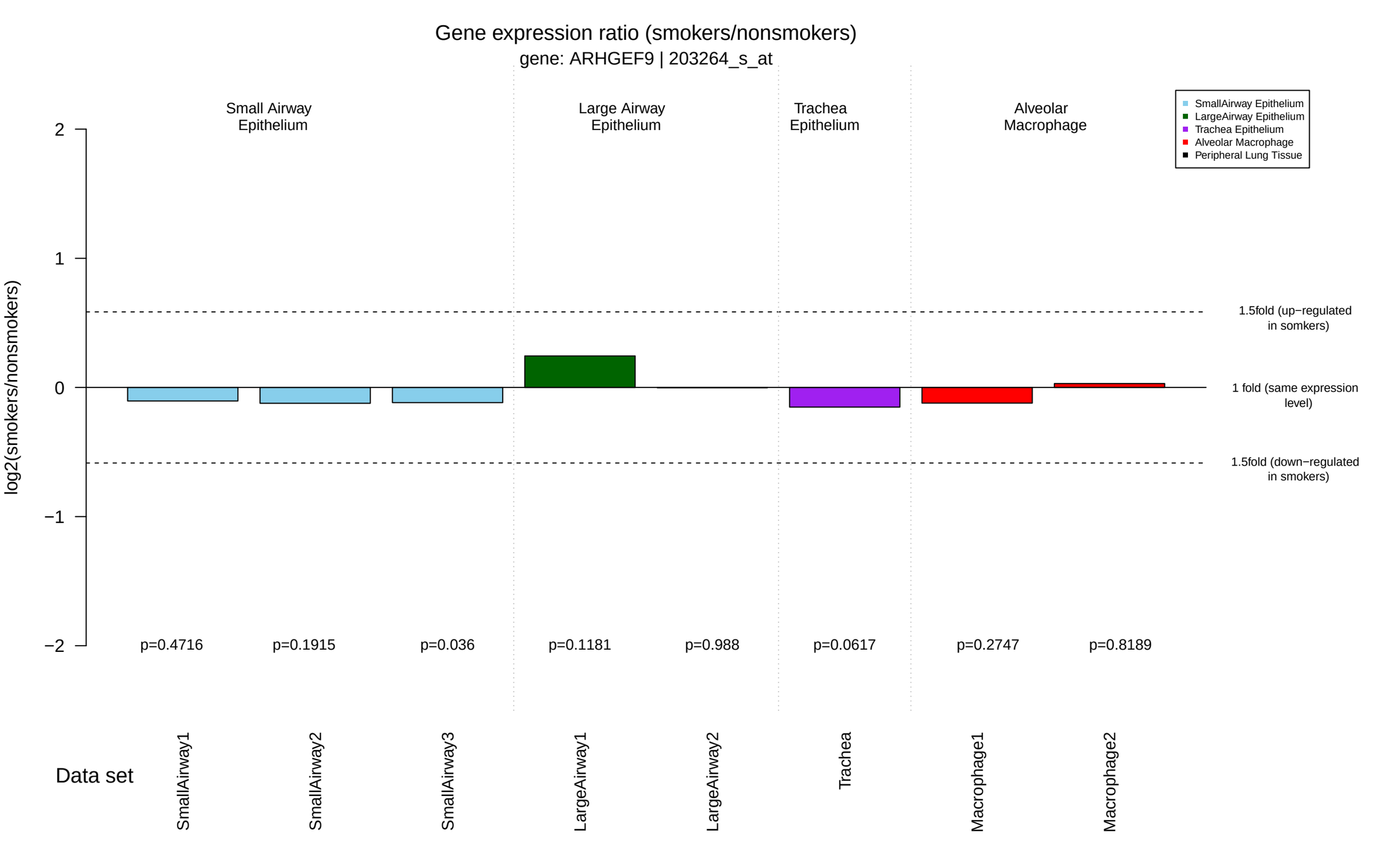
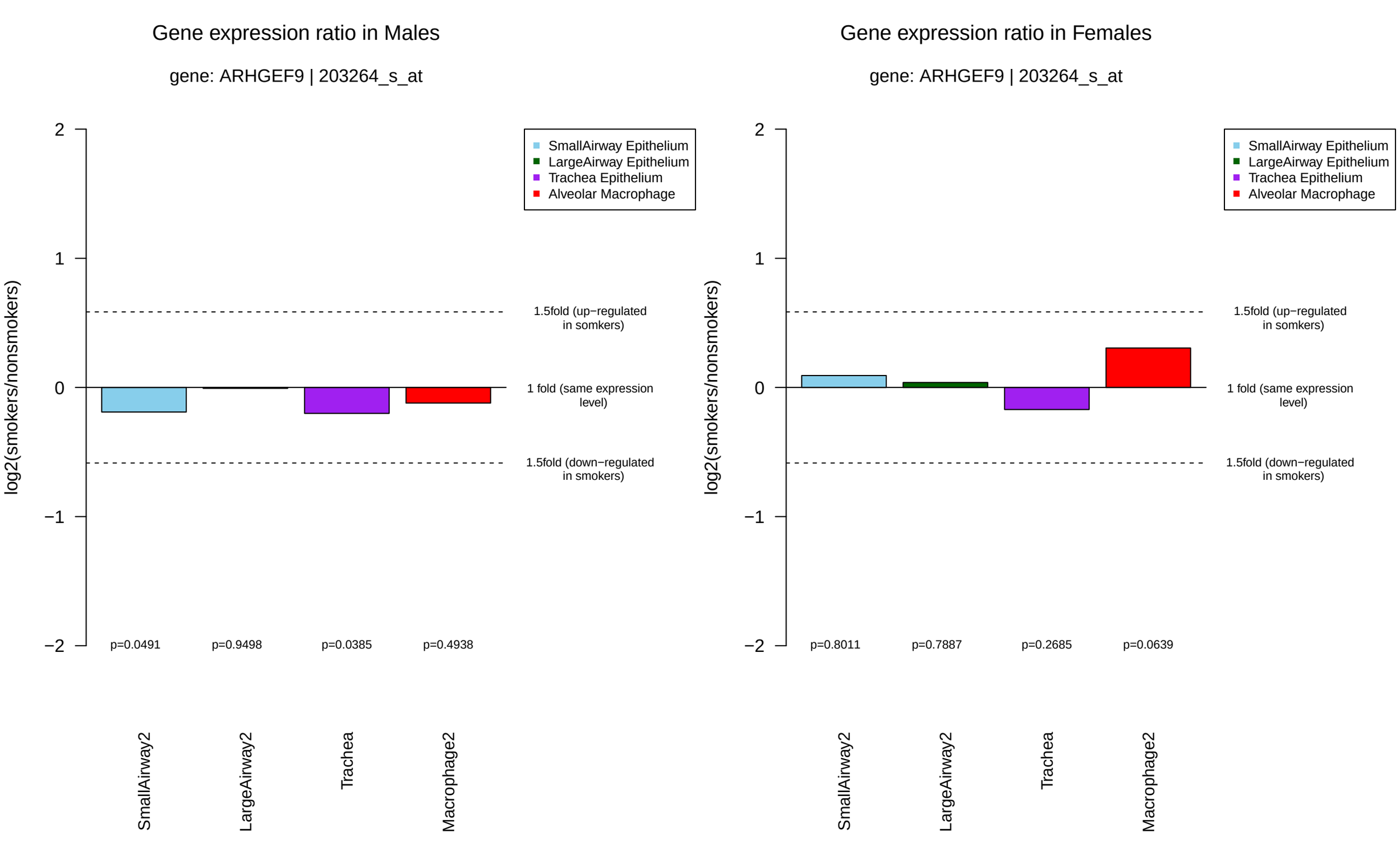
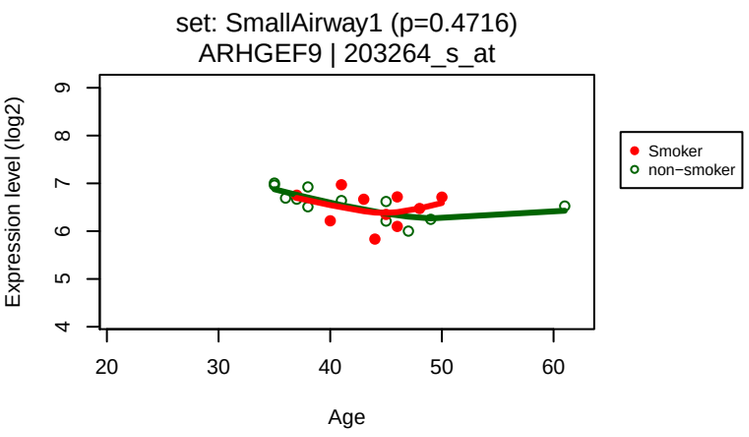
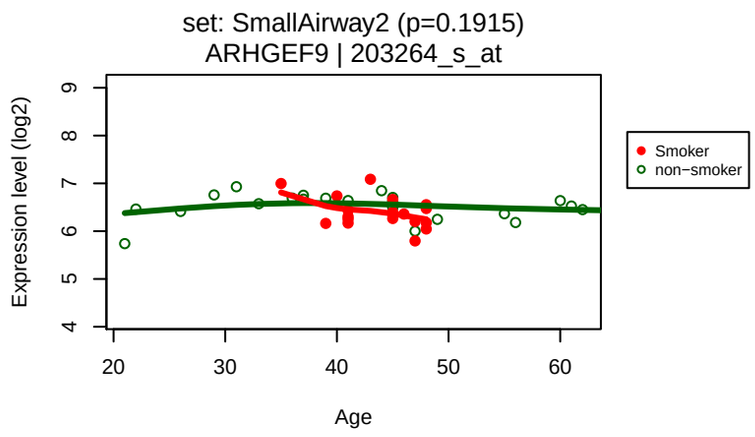
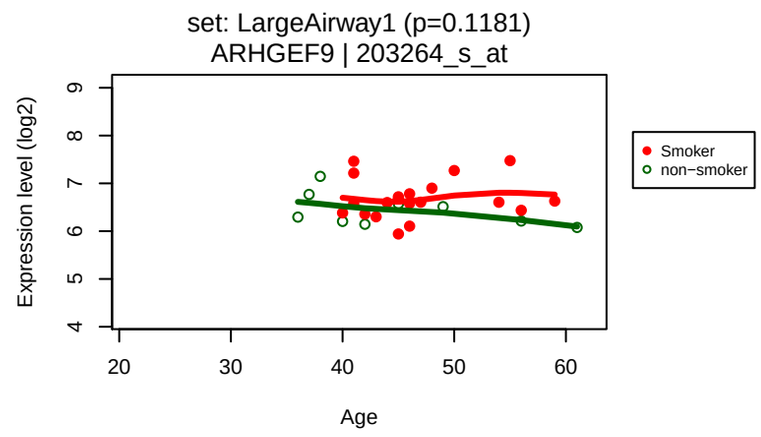

Gene expression of smokers and nonsmokers
gene: ARHGEF9 | 203264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4716
p=0.1915
p=0.036
p=0.1181
p=0.988
p=0.0617
p=0.2747
p=0.8189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

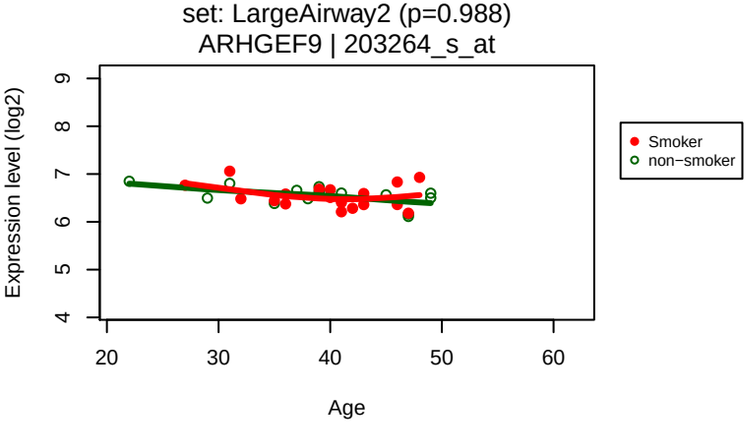

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF9 | 203264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4716
p=0.1915
p=0.036
p=0.1181
p=0.988
p=0.0617
p=0.2747
p=0.8189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

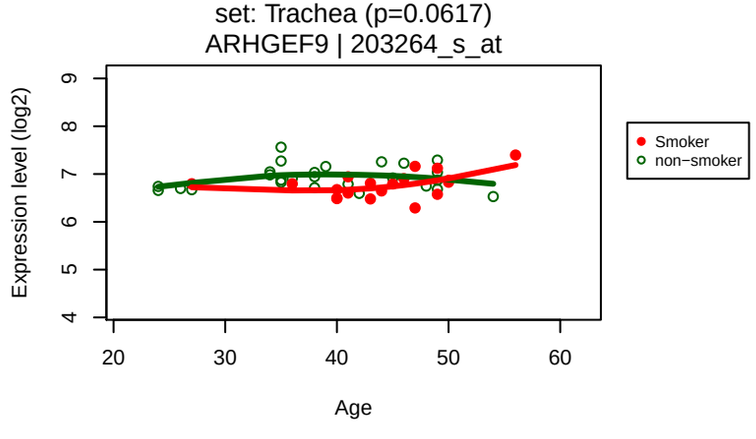

Gene expression ratio in Males
gene: ARHGEF9 | 203264_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0491
p=0.9498
p=0.0385
p=0.4938
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF9 | 203264_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8011
p=0.7887
p=0.2685
p=0.0639
SmallAirway2
LargeAirway2
Trachea
Macrophage2

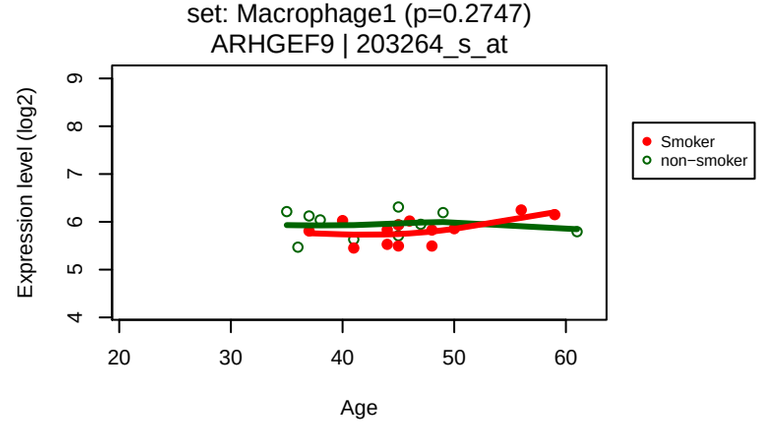

set: SmallAirway1 (p=0.4716)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker

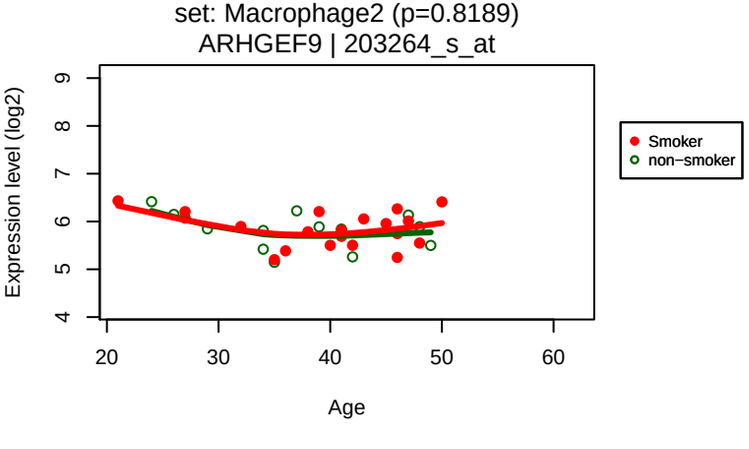

set: SmallAirway2 (p=0.1915)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker

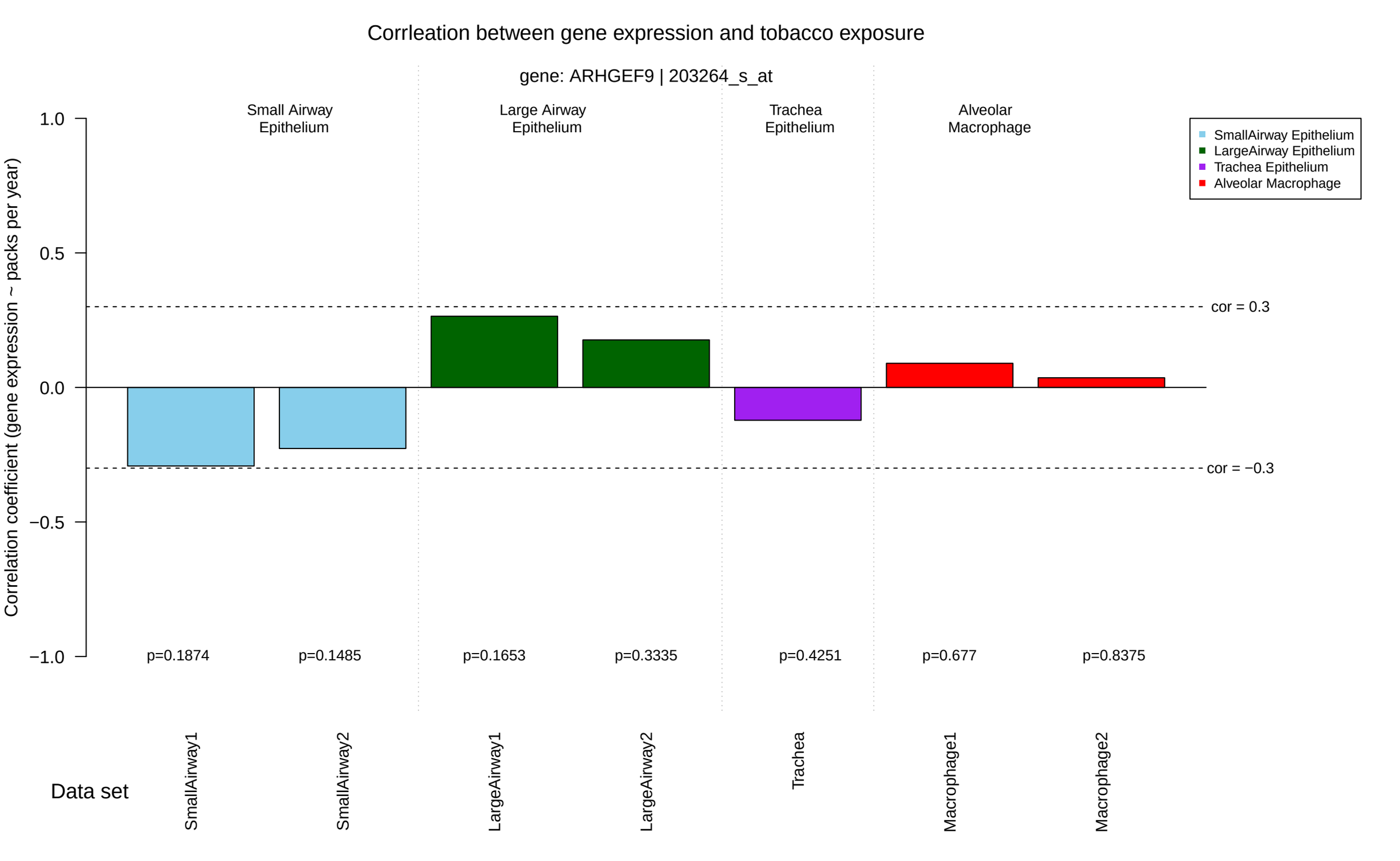

set: LargeAirway1 (p=0.1181)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.988)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0617)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2747)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8189)
ARHGEF9 | 203264_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF9 | 203264_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1874
p=0.1485
p=0.1653
p=0.3335
p=0.4251
p=0.677
p=0.8375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2